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H2**Started on** Tuesday, 25 August 2026, 5:03 PM**State** Finished**Completed on** Tuesday, 25 August 2026, 5:04 PM**Time taken** 36 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of $\text{arr}[i] * i$, where i is the index of the element ($i = 0, 1, 2, \dots, N$). Write an algorithm based on Greedy technique with a Complexity $O(n \log n)$.

Input Format:

First line specifies the number of elements- n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  // Comparison function for ascending order
5  int compare(const void *a, const void *b) {
6      long long x = *(const long long *)a;
7      long long y = *(const long long *)b;
8      if (x < y) return -1;
9      if (x > y) return 1;
10     return 0;
11 }
12
13 int main() {
14     int n;
15     if (scanf("%d", &n) != 1) return 0;
16
17     long long arr[n];
18     for (int i = 0; i < n; i++) {
19         scanf("%lld", &arr[i]);
20     }
21
22     // Sort array in ascending order (O(n log n))
23     qsort(arr, n, sizeof(long long), compare);
24
25     long long max_sum = 0;
26     for (int i = 0; i < n; i++) {
27         max_sum += arr[i] * i;
28     }
29
30     printf("%lld\n", max_sum);
31     return 0;
32 }
```

	Input	Expected	Got	
✓	5 2 5 3 4 0	40	40	✓

	Input	Expected	Got	
✓	10 2 2 2 4 4 3 3 5 5 5	191	191	✓
✓	2 45 3	45	45	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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